

Knowledge Injection Methods and Their Effect on Hallucination Rate on LLMs

Vasco Rodrigues¹

¹University of Coimbra, Department of Informatics Engineering
{vascorodrigues}@student.dei.uc.pt,

1 Introduction

Large Language Models (LLMs) are highly capable across general domains, yet their reliability decreases sharply in specialized areas such as scientific research. In these settings, LLMs frequently produce factually incorrect statements, hallucinations, which undermine their usefulness for tasks requiring precision, such as scientific information extraction. A common strategy to mitigate this issue is Knowledge Injection (KI), which aims to provide models with the external or domain-specific knowledge they lack. Among the available KI techniques, approaches such as Retrieval-Augmented Generation (RAG) and parameter-efficient fine-tuning (e.g., LoRA) have gained attention for their potential to ground LLM outputs and reduce hallucinations.

However, while KI methods are widely proposed as solutions, their effectiveness is not always systematically measured. This motivates a second line of research: hallucination quantification, which seeks to evaluate the factual correctness of LLM outputs using structured verification frameworks. Despite progress in both fields, direct comparisons of KI techniques under a unified hallucination-evaluation pipeline remain scarce, particularly in the scientific domain.

This project addresses this gap by evaluating how different KI methods influence hallucination rates during scientific triplet extraction. Specifically, we test two Qwen3 models (1.7B and 4B parameters) under three conditions: baseline (no KI), RAG, and LoRA. Using the SciERC dataset as a gold-standard Knowledge Graph and the RefChecker framework for hallucination verification, we aim to answer:

- **RQ1:** Do different KI methods lead to significantly different hallucination rates?
- **RQ2:** Can hallucination quantification serve as a reliable benchmark to compare KI strategies in scientific information extraction?

Our overall approach consists of extracting a gold-standard KG from SciERC, generating aligned questions, injecting knowledge through RAG or LoRA, prompting Qwen3 models to produce SciERC-style triplets, and evaluating their outputs using RefChecker. This allows for a controlled, quantitative comparison of KI methods and their impact on factual consistency in LLMs.

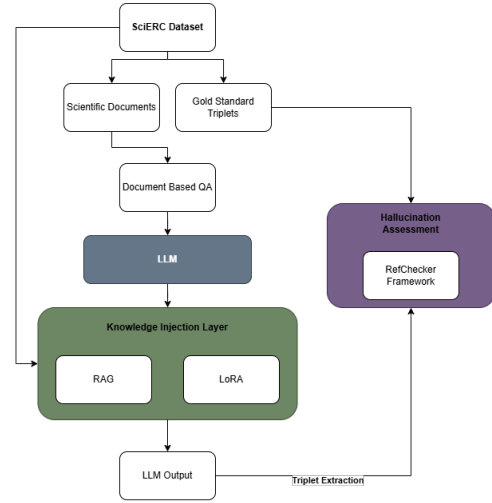


Figure 1: Pipeline

Every line of code can be consulted on the official github repository on Github-Repository

2 Related Work

2.1 Knowledge Injection into LLMs

LLMs have a static parametric memory, which limits their ability to incorporate new or domain-specific knowledge. This limitation has driven the development of multiple Knowledge Injection (KI) paradigms, including static parameter updates, retrieval-based approaches, hybrid architectures, and model-editing techniques. Below we summarize the main findings and limitations of the most relevant methods, also outlined in Table 1.

Static Knowledge Injection: Fine-Tuning

Ovadia et al. [Ovadia *et al.*, 2024] compare unsupervised Fine-Tuning (FT) with RAG, showing that RAG consistently outperforms FT across knowledge-intensive tasks, especially for injecting new information. FT proved unstable, highly sensitive to hyperparameters, and sometimes degraded performance. While data augmentation improved FT, the study’s generality is limited by small model sizes, a single knowledge source, and reliance on QA-based accuracy metrics.

Retrieval-Augmented Generation (RAG) and Parametric RAG (PRAG)

RAG [Lewis *et al.*, 2021] provides dynamic, context-time knowledge injection by retrieving external documents. It improves factuality but introduces latency and can hinder complex reasoning due to reliance on input-level context. PRAG [Su *et al.*, 2025] bridges retrieval and fine-tuning by encoding multiple representations of a document directly into the model’s FFNs. It improves parametric recall and reasoning without RAG’s runtime overhead. Limitations include high memory footprint and architecture dependence. PRAG complements rather than replaces RAG, as both can operate jointly.

Knowledge Model Editing (KME)

KME methods [Wang *et al.*, 2025] aim to update or correct model knowledge without full retraining. Approaches include:

- **External Memorization:** storing new knowledge in auxiliary memory or additional parameters;
- **Global Optimization:** constrained fine-tuning that integrates new knowledge while preserving prior capabilities;
- **Local Modification:** fine-grained edits to weights associated with specific facts.

Despite their promise, KME methods face challenges such as maintaining locality and generality simultaneously, preserving coherence across many cumulative edits, handling unstructured text, and supporting continual updates.

LoRA

LoRA [Hu *et al.*, 2021] introduces low-rank trainable matrices into frozen Transformer weights, enabling highly parameter-efficient fine-tuning. It reduces trainable parameters by several orders of magnitude and avoids additional inference latency, while allowing easy task-switching through adapter replacement. Experiments across RoBERTa, DeBERTa, GPT-2, and GPT-3 175B demonstrate performance comparable to or exceeding full fine-tuning, making LoRA a central tool in modern KI pipelines.

2.2 Hallucinations in LLMs

LLMs often produce content that is factually incorrect or unfaithful to source material [Farquhar *et al.*, 2024]. Their internal knowledge is limited by training data coverage and lack of post-training updates, making domain-specific tasks (e.g., science and medicine) particularly prone to hallucinations [Huang *et al.*, 2025]. KI methods, particularly RAG, help mitigate this by grounding outputs in external evidence.

Quantifying Hallucinations

Early hallucination detection relied on internal uncertainty estimators such as entropy [Farquhar *et al.*, 2024]. Later approaches adopted the *LLM-as-a-Judge* paradigm, where

an aligned LLM evaluates factuality [Huang *et al.*, 2025], though this raises bias and reliability concerns [Wang *et al.*, 2023].

Modern techniques compare model outputs against external references, distinguishing **factuality** (alignment with real-world facts) and **faithfulness** (alignment with context, instructions, or internal logic) [Huang *et al.*, 2025; ?].

Frameworks for Quantification

FactScore [Min *et al.*, 2023] decomposes outputs into atomic facts and checks each against a ground-truth source, directly measuring factual accuracy.

Ragas [Es *et al.*, 2025] evaluates RAG pipelines without a ground-truth KG, using metrics such as answer relevance, context relevance, and faithfulness—defined as the degree to which claims can be inferred from the provided context.

REFCHECKER [Hu *et al.*, 2024], the most relevant to our work, adopts a knowledge-centric approach by extracting claim-triplets and verifying each against a reference KG. Unlike most frameworks, it explicitly labels claims as **Supported**, **Contradicted**, or **Not Verifiable**, making it well-suited for scientific domains where information is often incomplete or ambiguous.

3 Curricular Unit Scope

This project fits the scope of the Knowledge and Language curricular unit because of this reasons:

- **Knowledge Representation:** It uses the Gold Knowledge Graph provided by the SciERC dataset as the factual reference for evaluating hallucinations in triplet extraction. The extracted triplets, composed of entities (e.g., Method, Task, Dataset) and relations (e.g., Used-For, Part-Of), are compared against this structured schema and classified as Supported, Contradicted, or Not Verifiable. By grounding evaluation in this triplet-based structure, the project directly engages with knowledge representation, as it relies on formalized entities, relations, and semantic links to assess the correctness of model-generated scientific knowledge.
- **Natural Language Processing:** The project uses LLMs (Qwen3 models) to: (1) Interpret scientific language, (2) Answer research questions, (3) perform information extraction, (4) Follow prompts, (5) generate triplets based on context or injected knowledge. Methods such as RAG, LoRA, embedding models, and triplet extraction are all core NLP techniques. Besides this the evaluation pipeline (RefChecker) also operates entirely on natural language outputs.

4 Approach

The literature review highlighted two core themes relevant to this project: (1) the growing importance of Knowledge Injection (KI) techniques such as RAG and parameter-efficient fine-tuning for grounding LLM outputs, and (2) the development of increasingly rigorous methods for quantifying hallucinations, particularly those based on knowledge verification

Table 1: Comparison of Knowledge Injection Paradigms

Paradigm	Mechanism	Advantages	Limitations
Static (FT)	Updates parameters directly	Persistent integration	Expensive; risk of forgetting
Static (LoRA)	Low-rank adapters on frozen weights	Efficient; no latency; modular	Limited capacity; task-specific
Dynamic (RAG)	Retrieval during inference	Up-to-date; flexible	Computational overhead
Hybrid (PRAG)	Parametric encoding of external knowledge	Strong recall and reasoning	High memory; architecture-tied
KME	Local/global post-training edits	Targeted; preserves base model	Scalability; consistency issues

rather than internal model heuristics. Although several works have evaluated KI strategies individually, direct comparisons of their impact on hallucination rates, especially within scientific information extraction tasks, remain limited. This gap is particularly relevant in specialized domains, where factual consistency is critical and model hallucinations can easily propagate erroneous scientific claims.

Building on this insight, the goal of this project is to systematically compare the effect of two KI methods, RAG and LoRA, on the ability of small and medium-sized LLMs to extract factual scientific triplets. The SciERC dataset provides an ideal benchmark framework, offering both a structured schema for triplet extraction and a “golden” Knowledge Graph (KG) that enables objective evaluation of correctness. To quantify hallucinations, we adopt the RefChecker methodology, which categorizes each predicted triplet as Supported, Contradicted, or Not Verifiable, providing a clear and interpretable measure of factual consistency.

With this in mind, the following pipeline was designed and implemented:

1. The reference KG was extracted directly from the SciERC dataset, providing the ground-truth set of triplets against which the models would later be evaluated.
2. A set of research questions was automatically generated from the same dataset by leveraging the sentence- and triplet-level structure of SciERC. This ensured full alignment between the prompts given to the LLMs and the information present in the gold standard.
3. Two KI techniques were applied:
 - (a) RAG: Document (Doc-1) was chunked and embedded; at inference time, the top 15 most similar chunks were retrieved and appended to each question, grounding the model’s reasoning.
 - (b) LoRA: The relevant scientific document was converted into a Hugging Face dataset and used to fine-tune Qwen3-4B via low-rank adapters, enabling parameter-level knowledge integration without modifying the base model. Baseline models without KI were also included for comparison.
4. LLMs were prompted using structured templates designed to enforce strict compliance with the SciERC

triplet format. RAG models received retrieved context; LoRA and baseline models received only the question.

5. Each model’s output went through sanitization and was then classified by the RefChecker into Supported, Contradicted, or Not Verifiable categories. Hallucination rate was defined as the proportion of Contradicted + Not Verifiable triplets. This enabled a direct comparison of the factual reliability of each KI method across different model sizes.

This approach provides a controlled experimental setup for assessing how different forms of knowledge injection influence hallucination rates in scientific information extraction. By grounding KI methods in a formal KG schema and evaluating their performance through a triplet-based hallucination framework, the project contributes a structured comparison of RAG and LoRA in a scientifically oriented task domain.

5 Dataset

The current study utilizes the SciERC (Scientific Information Extraction) benchmark dataset, which was specifically designed for information extraction tasks. This dataset contains a diverse collection of scientific papers along with the corresponding triplets that can be extracted from each document.

Due to its structured formalization, SciERC provides an ideal foundation for this research. The pre-extracted and validated relationships serve as the ground truth against which model outputs are evaluated.

The dataset comprises 106 scientific papers and their associated relationships. Each paper is segmented into chunks that are optimized for large language model (LLM) processing, making them particularly suitable for knowledge injection methods such as Retrieval-Augmented Generation (RAG), which is employed in this study. Each chunk corresponds to a sentence or phrase and includes the relationships that can be extracted from it.

These relationships are formalized as structured triplets:

[subject : label, relationship, object : label]

The label indicates the entity type of the subject or object and is drawn from a predefined set: Method, Task, Dataset

The relationship is also selected from a fixed ontology containing the following types: Used-For, Part-Of, Compare-With, SubClass-Of, Synonym-Of, Evaluated-With, Benchmark-For, Trained-With, SubTask-Of.

This well-structured formalization enables LLMs to be directed to output triplets in the prescribed format without requiring extensive output sanitization, thereby streamlining the evaluation process.

6 Pipeline Operation

6.1 Gathering of the Gold Knowledge Graph

The SciERC dataset contains annotated scientific papers, but due to time and computational constraints, the project operated on a reduced subset. From the 106 available papers, a single document was randomly selected using seed 42: doc_146120936 (referred to as Doc-1). Its extracted Knowledge Graph contains 80 nodes, 125 edges, and 71 chunks. In comparison, the full dataset KG (5,640 nodes and 12,083 edges) was far beyond feasible for this project.

Although SciERC is well structured, additional preprocessing was required. The document’s text was extracted into a separate file, and sentence-level triplets were consolidated into three broader contextual sections. This restructuring simplified vector-database queries and facilitated LoRA fine-tuning.

Questions were then generated from the gold-standard triplets using the template:

“A LABEL (subject) is RELATIONSHIP of the OBJECT”

For example, “Which method is used for object detection?” corresponds to the triplet:

[Subject:Method, Used-For, Object detection:Task]

This ensured alignment between the questions and the expected SciERC schema.

6.2 Experimental Procedure

The SciERC dataset provides papers, sentence-level text, and labeled triplets. For this project, questions were derived from each sentence or triplet set so that models could attempt to reproduce SciERC-compliant triplets. Only content from Doc-1 was used in the gold standard to avoid evaluation bias. Clear, consistent prompting was required to ensure reliable triplet extraction.

6.3 Models Used

Two models from the Qwen3 family were selected: Qwen3-1.7B and Qwen3-4B. These models were chosen due to improvements in reasoning, STEM coverage, and instruction-following. Earlier Qwen2.5 models were tested but proved unreliable, often failing to follow strict output formats.

Two Knowledge Injection methods were applied: RAG and LoRA. Due to tool support differences, RAG was implemented using Qwen3 models running on Ollama, while LoRA fine-tuning was carried out via Hugging Face. PRAG

was considered but discarded due to hardware limitations and compatibility issues.

This hybrid setup introduced some constraints: Hugging Face does not support Qwen3-1.7B, limiting LoRA evaluation to Qwen3-4B. Furthermore, notable behavioral differences emerged between model distributions. Ollama-served models consistently followed instructions more reliably due to:

- curated system prompts and chat templates,
- stable decoding defaults optimized for deterministic output,
- GGUF quantization and runtime optimizations.

In contrast, Hugging Face models required manual prompt formatting and exhibited greater variability across inference environments.

Given these conditions, the following model variants were evaluated:

- **Qwen3-1.7B:** Baseline and RAG.
- **Qwen3-4B:** Baseline, RAG, and LoRA.

Each model was compared in terms of hallucination rate and distribution of Supported, Contradicted, and Not Verifiable triplets.

6.4 Prompts

As mentioned previously, clear and specific instructions are essential for accurate triplet extraction from the LLM. To this end, a structured prompt template was designed for use with both model types.

This prompt is organized into five core sections to ensure reliable and structured output, particularly for the RAG approach:

- **Task:** Formalizes the LLM’s role as an information extraction system. It instructs the model to answer a scientific question using the provided context, enumerates the allowed entity and relationship labels, and defines strict output rules to prevent deviation from the SciERC format.
- **Question:** Explicitly marks where the question appears in the prompt, ensuring the LLM identifies it correctly without confusion.
- **Context:** Exclusive to RAG, this section supplies the retrieved document chunks most similar to the question, serving as the knowledge source for triplet extraction.
- **Output Format:** Enforces the required JSON structure for the LLM’s response, mandating a valid JSON object containing only the "triplets" array with no additional text.
- **Example:** Provides a concrete demonstration of an input context and its corresponding output in SciERC-compliant format, helping smaller models adhere more closely to instructions.

For LoRA, a modified prompt was employed for two main

reasons: to ensure a fair comparison without injected context, and to accommodate the distribution method of Hugging Face models. This version is intentionally simpler and retains only three sections:

- **Task:** Describes the LLM’s role as an information extraction system, specifying that it must output factual triplets in JSON format given a research question. The same labeling rules apply, but without any retrieved context, the model relies solely on knowledge from its trained LoRA adapter.
- **Question:** As in the RAG prompt, this section clearly demarcates the question to avoid ambiguity.
- **Output Format:** Enforces the required JSON structure, maintaining strict adherence to the SciERC triplet format.

The Example section was omitted because it tended to overwhelm the model, often causing it to reproduce the example verbatim across different questions. The Context section was removed to ensure a fair comparison between knowledge injection methods, requiring the LoRA model to depend exclusively on its fine-tuned adapter for knowledge recall.

6.5 RAG Operation

RAG was the first Knowledge Injection method employed in this project. The selected document (Doc-1) was divided into sentence-level chunks, each converted into an embedding and stored in a local JSON-based vector database. During inference, both the question and document chunks were embedded into the same vector space. Cosine similarity was used to retrieve the top 15 most relevant chunks, which were then appended to the question. This grounded the LLM’s reasoning in the source document, reducing hallucinations and increasing the likelihood of producing accurate SciERC-compliant triplets.

6.6 LoRA Operation

LoRA served as the second Knowledge Injection technique, offering a static alternative to RAG. Instead of modifying the full model, LoRA freezes all original weights and introduces trainable low-rank matrices that efficiently capture new knowledge. This avoids catastrophic forgetting and substantially reduces computational cost.

The scientific document used for KI was transformed into a Hugging Face dataset, tokenized, padded with a custom `<|pad|>` token, concatenated, and segmented into fixed-length training blocks suitable for causal language modeling. The Qwen3-4B model was loaded and resized to include the new token, after which LoRA adapters were attached to key transformer modules.

LoRA was configured with rank 8, scaling factor 2, and dropout 0.1, balancing expressivity and regularization. After training, only the lightweight adapter weights were saved. These adapters were then loaded (without merging) using the PEFT framework, ensuring the base model remained unchanged. Each question was subsequently answered without context, enabling direct measurement of how much knowl-

edge the model had internalized through LoRA alone.

6.7 RefChecker Operation

RefChecker was applied as the final evaluation component for measuring hallucinations under the two KI strategies. Since our setup compares LLM-generated triplets to a gold-standard KG, the project fits RefChecker’s “QA with reliable context” scenario. The framework consists of two stages, but one more was added because it presented more clarity for the redaction of the paper: Claim Extractor, Hallucination Verifier, and Aggregator.

First Stage: Claim Extractor

This stage extracts factual claims as triplets. In our case, extraction is inherent to the task, since the LLM is explicitly prompted to output SciERC-formatted triplets. Each triplet therefore becomes an individual claim to be verified.

Second Stage: Hallucination Verifier

Each predicted triplet is compared against the gold-standard KG. Due to budget limitations, commercial judge models such as GPT-4 or Claude 2 could not be used. Instead, the free-tier GROQ API (`groq/compoud` model) served as the judge. It received the gold-standard triplet, the question, and the LLM output, and returned a numerical label: 0 (Supported), 1 (Contradicted), or 2 (Not Verifiable). The model followed instructions reliably, producing clean categorical outputs in all cases.

Third Stage: Aggregator

RefChecker aggregates the judged triplets into evaluation metrics. For each model, we computed:

- Supported rate: $\frac{\text{Supported}}{\text{total}} \times 100$,
- Hallucination rate: $\frac{\text{Contradicted} + \text{Not Verifiable}}{\text{total}} \times 100$.

Each question was manually aligned with its corresponding gold-standard triplet from SciERC to guarantee accurate evaluation.

7 Results

Following output sanitization and the second stage of RefChecker evaluation, the results from each model were aggregated for comparative analysis. In addition to testing models enhanced with Knowledge Injection (KI) methods, baseline models without KI were also evaluated through the same triplet extraction pipeline. This provided a reference point for assessing the impact of KI techniques.

Comparing KI-enhanced models against their baseline counterparts reveals several key findings:

- For the 1.7 models, RAG implementation yielded a measurable improvement in hallucination rates. Although only RAG could be tested at this scale, as Hugging Face does not support Qwen3-1.7B models, and Qwen2.5-1.5B performed poorly in following strict formatting rules, the results were consistent with expectations. The baseline 1.7B model exhibited a hallucination rate of 91.8%, while the RAG-enhanced version reduced this to 75.5%. This supports the conclusion that injecting

knowledge at the context level effectively reduces hallucinations in LLM outputs.

- For the 4B models, both KI methods, LoRA and RAG, improved performance relative to the baseline. The base 4B model hallucinated 95.9% of triplets, compared to 79.6% for LoRA and 49.0% for RAG. While LoRA's improvement was modest and similar in magnitude to RAG's impact on the 1.7B model, RAG on the 4B model proved substantially more effective. This suggests that larger models are better equipped to extract structured triplets from contextual information. LoRA's limited gains may stem from the small training corpus, only one relevant paper was available for fine-tuning, which may have been insufficient for the adapter to internalize robust domain knowledge.

Overall, RAG on the 1.7B model reduced hallucination by 17.8%; LoRA on the 4B model reduced it by 17.0%; and RAG on the 4B model produced the largest improvement, cutting hallucination by 48.9% relative to the baseline. This establishes RAG, particularly with larger models, as a reliable and potent KI method.

It was also verified that the 1.7B model had a lower hallucination percentage than the 4B model, a difference of 4.1%. This could indicate that a greater number of parameters does not invariably produce a more factually accurate model. Alternatively, given the difference is not substantial, it may simply reflect the inherent randomness in the models' outputs rather than a systematic performance advantage.

It's possible to see that most of the models answers, were not contradicted, but yes not verifiable, this happened because of the way prompts were done. It was always enforced that the model should output the a JSON in the format of:

```
"triplets": [
  "[Entity:LABEL, Relationship, Entity:LABEL]"
]
```

This combined with the fact that the model was ask to try to produce factual triplets most of the times, when not knowing the answer, the model simply outputted:

```
"triplets": [
  "[]"
]
or
"triplets": [
  "[:Method, Part-Of, :Method]"
]
```

This would strictly follow the rules given and would not be classified as a contradictions simply not verifiable. To try and bridge this the model was proposed to try and Infer triplets from its knowledge or the context given, while not inventing new labels for the entities and relationships, so that he would continue to follow the SciERC format. This did decreased the number of this occurrences and more concrete hallucinations appeared. So the source of not verifiable triplets is: (1) empty triplets, (2) triplets containing

only label/relationship templates, and (3) triplets unverifiable against the gold standard or provided context given to the Judging LLM.

Following the RefChecker framework, hallucinations were defined as the sum of contradicted and not verifiable triplets, explaining the high overall hallucination rates. Breaking down the results by KI method:

- For LoRA, in spite of the new knowledge introduced by the trained adapter the model still had more contradicted triplets than supported. Still the most classified category was Not Verifiable, either because it didn't knew the answer and outputted a empty triplet or because it gave a triplet that could not be verified with the given context.
- For RAG on the 1.7B model there were more Supported triplets then Contradicted, still there were a lot of Not Verifiable triplets even more then the ones introduced by LoRA.
- For RAG on the 4B model we could see a major improvement, the most categorized type was Supported, with fewer Not Verifiable and Contradicted results, this was the model combination with the best performing results (most Supported triplets), least not verifiable and contradicted answers.

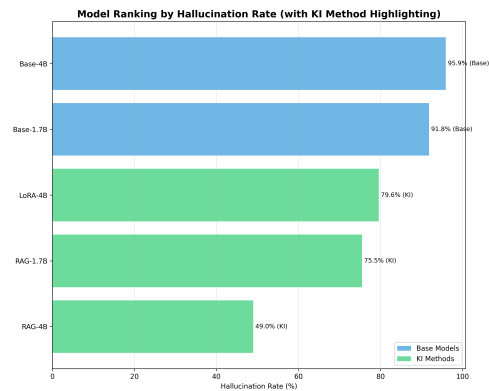


Figure 2: Model ranking by hallucination rate with Knowledge Injection highlighting

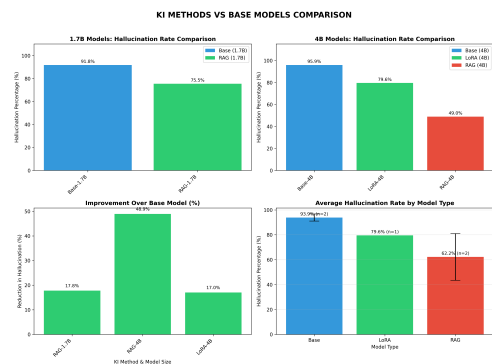


Figure 3: Comparison of Knowledge Injection methods versus base models

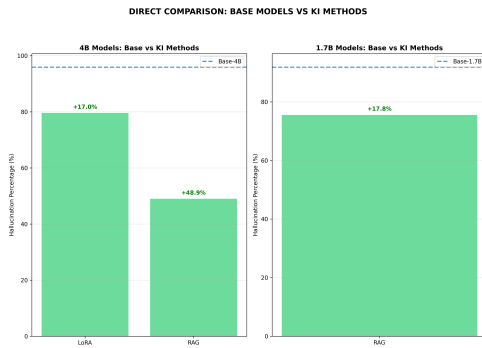


Figure 4: Direct comparison between base models and Knowledge Injection methods

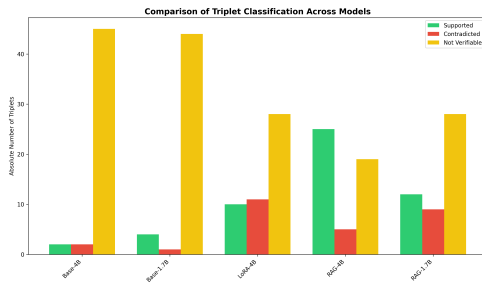


Figure 5: Absolute triplet comparison across models

8 Discussion

The experiments evaluated two Knowledge Injection (KI) methods, RAG and LoRA, across two model sizes, Qwen3-1.7B and Qwen3-4B, with the goal of measuring their impact on factuality and hallucination reduction under the SciERC schema. This section summarizes the main findings, their implications, and the limitations observed.

8.1 Impact of Knowledge Injection on Hallucination Reduction

All KI methods improved performance relative to baseline models, though to different degrees. RAG provided the strongest gains, especially for Qwen3-4B, reducing hallucinations from 95.9% to 49%. This demonstrates that larger models effectively leverage retrieved context and benefit substantially from dynamic grounding.

LoRA also reduced hallucinations (95.9% to 79.6%), but the improvement was smaller. Since the adapter was trained on only one scientific paper, the limited corpus likely constrained its ability to internalize domain knowledge, leading to inconsistent triplet generation. A larger fine-tuning dataset would likely increase LoRA’s effectiveness.

For Qwen3-1.7B, RAG yielded a modest reduction (91.8% to 75.5%), showing that smaller models do benefit from context injection, though less than their larger counterparts. Overall, model size moderates the effectiveness of KI: the same context yields much stronger fact-grounding in the 4B model than in the 1.7B model.

8.2 Relative Performance of Model Sizes

The baseline Qwen3-4B model exhibited slightly higher hallucination rates than the 1.7B model (95.9% vs. 91.8%), indicating that larger models are not inherently more factual. This may stem from greater output verbosity or simple random variation, as the 4.1-point difference is small. Under KI conditions, however, the 4B model consistently outperformed the 1.7B model, confirming that parameter count becomes advantageous once relevant knowledge is provided.

8.3 Sources of Not Verifiable Triplets

A large share of hallucinations fell into the “Not Verifiable” category. This occurred because:

- the models sometimes generated empty or template-like triplets to satisfy strict formatting constraints,
- inferred relations were not always present in the gold standard,
- the enforced JSON-only structure reduced contradictions but increased unverifiable outputs.

These results highlight the challenge of balancing rigid output schemas with the limited reasoning abilities of smaller models. When overly constrained, models default to placeholders instead of producing meaningful content.

8.4 RAG vs. LoRA

The comparison between the two KI strategies is summarized in Table 2. RAG emerged as the most effective approach, particularly for the 4B model. LoRA remained attractive due to its modularity and efficiency, but with such a small fine-tuning corpus, it lacked the breadth necessary to match RAG’s performance.

Method	Strengths	Weaknesses
RAG	Strong grounding in source text; largest hallucination reduction; effective at all model scales	Requires retrieval infrastructure; quality depends on chunking and embeddings
LoRA	Lightweight; no context required at inference; efficient and modular	Limited by size of fine-tuning corpus; struggles with small datasets

Table 2: Comparison of Knowledge Injection Methods

8.5 Model Distribution Differences (Ollama vs. Hugging Face)

Finally, the study revealed differences between Ollama-distributed and Hugging Face-distributed Qwen models. Ollama versions exhibited better instruction-following behavior due to:

- curated system prompts and chat templates,
- standardized decoding defaults,
- GGUF quantization and runtime optimizations.

Hugging Face versions required more manual configuration and were more prone to schema deviations, which complicated LoRA evaluation. These results demonstrate that model behavior depends not only on weights, but also on the inference environment and distribution.

8.6 Answering the Main Questions

Having all this information in account we can finally answer our initial research questions:

RQ1 - Do different KI methods lead to significantly different hallucination rates?

Short answer yes. The results presented above show clear and substantial differences between the KI methods: (1) RAG (4B) delivered the strongest improvement, cutting hallucinations from 95.9% to 49.0% (a 48.9% reduction); (2) RAG (1.7B) also improved performance, though less dramatically from 91.9% to 75.5% (a 17.8% reduction); LoRA (4B) reduced hallucinations from 95.9% to 79.6% (a 17.0% reduction)

These differences are sufficiently large and systematic to conclude that the use of a KI method has a significant impact on hallucination rates, specially when comparing RAG to LoRA and baseline models. But it is also possible to verify that the setup of the problem must be taken into account when choosing these KI methods, with a much larger corpus LoRA would have represented data much better on its adapter, while RAG would become increasingly slower because of the sheer number of similarity calculations it had to do.

Thus, different KI methods do not perform equally, and their impact on hallucinations is both measurable and method and environment dependent.

RQ2 - Can hallucination quantification serve as a reliable benchmark to compare KI strategies in scientific information extraction.

Short answer yes. The results demonstrated the hallucination quantification, implemented through the RefChecker framework, provided a stable, discriminative, and interpretable benchmark for the comparison of our KI methods.

The framework differentiated clearly between Supported, Contradicted, and Not Verifiable triplets. It exposed meaningful differences between methods:

- RAG (4B) produced the most Supported and fewest hallucinated triplets.
- LoRA showed moderate improvement but still struggled to generate verifiable relations
- Baseline models performed consistently worse

Therefore, hallucination quantification proved reliable and well-suited to be a benchmarking tool for evaluating Knowledge Injection methods in scientific information extraction.

Although this is true, some adjustments must be done to RefChecker's like framework since it is still difficult to differentiate between hallucinated outputs or default placeholders

on the case of "Not Verifiable" categorized triplets.

9 Conclusion

This study evaluated the performance of Qwen3 models under Knowledge Injection using two techniques, RAG and LoRA, on a structured triplet extraction task defined by SciERC. The findings demonstrate that Knowledge Injection substantially reduces hallucination rates and improves factual output quality, with RAG consistently outperforming LoRA across all configurations.

The primary conclusions are:

1. RAG was the most effective Knowledge Injection method, especially on the 4B model, achieving the highest number of Supported triplets and the lowest hallucination rate of 49%
2. LoRA presented moderate improvements, this happened because of the limited training corpus; with larger datasets, it may bridge part of the gap with RAG.
3. Model size does matter, but only when paired with appropriate Knowledge Injection methods; baseline 4B performance was unexpectedly worse than 1.7B, yet improved dramatically under the KI setup
4. Strict formatting requirements drive large proportions of "Not Verifiable" triplets, emphasizing the need for balanced prompt design.
5. Serving environments affect evaluation, Ollama model presented stronger instruction-following capabilities than equivalent Hugging Face models, influencing comparative Knowledge Injection performance.

Overall, the results support the conclusion that context-grounded Knowledge Injection (RAG) is the most reliable strategy for reducing hallucinations in small to mid-scale open-source LLMs, this when there isn't much information available. However, adapter-based Knowledge Injection (LoRA) remains a valuable approach when inference-time context is unavailable, offering lightweight, modular specialization.

Besides this this work proves that Knowledge Injection methods have significantly different performance results, and when choosing one the environment and experimental setup must be taken into account. Furthermore Hallucination quantification proved to be a reliable and well suited benchmarking tool for comparing these Knowledge Injection techniques in the context of scientific information triplet extraction.

Future work should include scaling LoRA training to a larger corpus, and experiment with hybrid Knowledge Injection methods, such as implementing RAG with a LoRA adapter, and evaluate RefChecker-like pipelines to better distinguish between true hallucinations and unverifiable outputs due to schema constraints.

References

[Es *et al.*, 2025] Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. Ragas: Automated Eval-

uation of Retrieval Augmented Generation, April 2025.

Computing Surveys, 57(3):1–37, March 2025.

- [Farquhar *et al.*, 2024] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, June 2024.
- [Hu *et al.*, 2021] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models, October 2021. arXiv:2106.09685 [cs].
- [Hu *et al.*, 2024] Xiangkun Hu, Dongyu Ru, Lin Qiu, Qipeng Guo, Tianhang Zhang, Yang Xu, Yun Luo, Pengfei Liu, Yue Zhang, and Zheng Zhang. Knowledge-Centric Hallucination Detection. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6953–6975, Miami, Florida, USA, November 2024. Association for Computational Linguistics.
- [Huang *et al.*, 2025] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. *ACM Transactions on Information Systems*, 43(2):1–55, March 2025.
- [Lewis *et al.*, 2021] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, April 2021.
- [Min *et al.*, 2023] Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation, October 2023.
- [Ovadia *et al.*, 2024] Oded Ovadia, Menachem Brief, Moshik Mishaelli, and Oren Elisha. Fine-Tuning or Retrieval? Comparing Knowledge Injection in LLMs, January 2024. arXiv:2312.05934 [cs].
- [Su *et al.*, 2025] Weihang Su, Yichen Tang, Qingyao Ai, Junxi Yan, Changyue Wang, Hongning Wang, Ziyi Ye, Yujia Zhou, and Yiqun Liu. Parametric Retrieval Augmented Generation, January 2025. arXiv:2501.15915 [cs].
- [Wang *et al.*, 2023] Cunxiang Wang, Xiaoze Liu, Yuanhao Yue, Xiangru Tang, Tianhang Zhang, Cheng Jiayang, Yunzhi Yao, Wenyang Gao, Xuming Hu, Zehan Qi, Yidong Wang, Linyi Yang, Jindong Wang, Xing Xie, Zheng Zhang, and Yue Zhang. Survey on Factuality in Large Language Models: Knowledge, Retrieval and Domain-Specificity, December 2023.
- [Wang *et al.*, 2025] Song Wang, Yaochen Zhu, Haochen Liu, Zaiyi Zheng, Chen Chen, and Jundong Li. Knowledge Editing for Large Language Models: A Survey. *ACM*